

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm. Examination

University Theory Examination November 2016

PH317 Pharmaceutical Analysis I

Date: 25.11.2016, Friday

Time: 10:00a.m. to 1:00p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Indian Pharmacopoeia Commission (IPC) is an Autonomous Institution of the
[A] Ministry of Health and Family Welfare, Govt. of India.
[B] Ministry of Health and Individual Welfare, Govt. of India.
[C] Ministry of External Affairs and Welfare, Govt. of India.
[D] Ministry of Medicine, Govt. of India.
 2. Buffer solution is made up of
[A] strong acid with a salt
[B] weak acid with a salt
[C] weak acid and strong acid
[D] mixture of salts
 3. Equivalent weight of H_2SO_4 is (Mol. Wt.=98g/mole)
[A] 98
[B] 49
[C] 32.66
[D] 196
 4. Weak Electrolyte is a substance which
[A] dissociates completely
[B] does not dissociate
[C] dissociates partially
[D] none of the above.
 5. Indicator range for methyl orange is
[A] 2-6
[B] 1-3
[C] 3-5
[D] 4-7
 6. Karl fisher titration is used for determination of
[A] Water content
[B] temporary hardness of water
[C] permanent Hardness of Water

- [D] Water content and hardness of water
7. The assay of a Calcium containing tablet is performed by
[A] Argentometric titration
[B] Non-aqueous titration
[C] Precipitation titration
[D] Complexometric titration
8. Commonly used titrant for non-aqueous titration is
[A] Benzoic acid
[B] EDTA
[C] Perchloric Acid
[D] AgNO_3
9. Indicator used in Fajan's Precipitation titration is
[A] Potassium Chromate
[B] Ferric alum
[C] Fluorescein
[D] Mordant Black II
10. Crystal Hydrates are used for _____ of Karl Fisher reagent
[A] Storage
[B] Stabilization
[C] Solubilization
[D] Standardization
11. The dielectric constant of water at 14.5 degree Celsius is _____
Calorie
[A] 0.5
[B] 1.0
[C] 1.5
[D] 2.0
12. What indicators are appropriate for complexometric titrations?
1. potassium chromate
2. eriochrome black T
3. murexide
4. methyl red
[A] 2,3
[B] 1,2,3
[C] 2,3,4
[D] 2,4
13. Procedure involved in Gravimetry is
[A] Preparation of sample, Precipitation, Digestion, Filtration, Washing, Drying, Weighing
[B] Preparation of sample, Precipitation, Filtration, Digestion, Washing, Drying, Weighing
[C] Preparation of sample, Filtration, Precipitation, Digestion, Washing, Drying, Weighing
[D] Precipitation, Digestion, Filtration, Washing, Drying, Preparation of sample, Preparation of sample, Weighing
14. Percentage Volume/ Volume means
[A] volume of solute in 1000ml solvent
[B] volume of solute in 100ml solvent
[C] volume of solute in 100gm of solvent
[D] gm of solute in 100ml of solvent

15. The addition of hydrofluoric acid and _____ to water produces a buffer solution.
 [A] HCl
 [B] HF
 [C] NaF
 [D] NaBr
16. According to _____ principle, if a reaction in equilibrium is subjected to a change, the reaction tends to move in such a direction that the effect of the change would be neutralized.
 [A] Le Chateliers Principle
 [B] Common ion Effect
 [C] Buffer action
 [D] Degree of dissociation
17. pH of water is 7. When a substance Y is dissolved in water, the pH becomes 2. The substance Y is a salt of
 [A] weak acid and weak base
 [B] strong acid and strong base
 [C] weak acid and strong base
 [D] Strong acid and weak base
18. pOH value means
 [A] negative log of Hydrogen ions
 [B] positive log of hydrogen ions
 [C] negative log of hydroxyl ions
 [D] positive log of hydroxyl ions
19. Sulphuric acid is
 [A] Non-electrolyte
 [B] Strong electrolyte
 [C] Weak Electrolyte
 [D] None of the above
20. Sampling errors can be minimized by
 [A] sampling by different analyst
 [B] increasing sample size
 [C] sampling multiple times
 [D] sampling at regular time interval

SECTION – II

- Q 2** Attempt any **TWO** of the following
- | | | |
|----------|--|-----------|
| A | Explain Buffer Action with suitable examples. | 05 |
| B | Write a note on contents of a monograph as per Indian Pharmacopoeia. | 05 |
| C | Explain the factors affecting Precipitation Titration. | 05 |
- Q 3** Attempt any **TWO** of the following
- | | | |
|----------|---|-----------|
| A | Derive Henderson Hasselbalch Equation. | 05 |
| B | i) Construct the gravimetric factor for estimation of precipitated NaCl (Mol. Wt. = 58.44g/mole) from AgNO ₃ (Mol. Wt. = 169.87g). | 02 |
| | ii) Calculate molarity of solution containing 50g K ₂ CO ₃ in 500ml. (Mol. Wt. = 138g/mole). | 02 |
| | iii) State the Molarity of 0.025N NaOH solution. | 01 |
| C | i) Calculate the pH and pOH of a solution containing 0.0005g ion per litre [H ⁺]. | 02 |
| | ii) Calculate the concentration of AgCl ions in a saturated AgCl solution at 25°C. K _{sp} of AgCl at 25°C is 1.8×10^{-10} . | 03 |

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Enlist the endpoint detection method in argentometric titration. Describe Mohr's method in detail. **05**
 - B** Write the principle used behind complexometric titration? Define Ligands and classify it. **05**
 - C** What is gravimetric analysis? Explain advantage, disadvantage and application of gravimetric analysis. **05**
 - D** Define Errors and explain classification of errors and minimization of errors. **05**
 - E** What is sampling? Explain sampling techniques of drugs. **05**
 - F** What is Hardness of water? Write a note on methods to remove hardness of water. **05**
- Q 5** Attempt any **FOUR** of the following
- A** Write a short note Karl Fisher titration. **05**
 - B** Write note on different types of complexometric titration. **05**
 - C** What is redox titration? Differentiate iodometry and iodimetry. **05**
 - D** Define neutralization curve and derive a curve for titration of strong acid and strong base. **05**
 - E** Discuss about co-precipitation and post precipitation. **05**
 - F** Write law of mass action and pharmacopoeial applications of Acid Base Titrations. **05**
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